

SUMMARY

Research and Developer with 1 year of hands-on experience in ERP system development, industrial automation, CNC machine integration, and web application development. Experienced in real-time MQTT communication, database integration, and report generation using Crystal Reports and HTML reporting. Proficient in C#, ASP.NET Core, ASP.NET MVC, JavaScript, HTML, CSS, and SQL Server, with a strong understanding of Object-Oriented Programming and MVC architecture. Skilled in developing responsive web applications, enhancing user experience, optimizing application performance, and delivering reliable software solutions in manufacturing and enterprise environments.

PROFESSIONAL EXPERIENCE:

Organization : Freedom Software solution
Designation : Research and Developer
Experience : 1 Year

SKILLS

- **Programming Language:** C#.
- **Framework:** ASP.NET Core, MVC, Entity Framework Core
- **Frontend:** HTML, CSS, JavaScript.
- **Database:** Microsoft SQL Server.
- **Web Development:** Responsive Web Design, UI/UX Design, Cross-Browser Compatibility, SEO Fundamentals
- **Industrial Integration:** MQTT Protocol, RS232 Communication
- **Concepts:** OOP, MVC Architecture, LINQ, Dependency Injection
- **Additional Knowledge:** Python, NumPy, Pandas

PROJECTS

1. CNC Machine Data Acquisition and Production Automation , Veltech Auto Part India Private Limited

- Developed a real-time data integration system to automate ERP production entry by collecting machine-level data from CNC and Fanuc machines.
- Retrieved live machine data such as running part details, process status, and cycle time for accurate production tracking.
- Built a backend application using C# to collect, process, and store machine data in a centralized database.
- Integrated Ethernet-enabled machines using IP-based communication for real-time data acquisition.
- Designed and implemented connectivity for legacy machines using RS232 communication, including DB25 to DB9 pin configuration and manual soldering of null modem cables.
- Configured Serial Device Servers (RS232 to Ethernet/Wi-Fi converters) for network-based data transmission.
- Automated manual production entry by directly syncing machine data with ERP modules, improving accuracy and reducing manual effort.
- Visited client locations and deployed the solution, ensuring proper integration between machines and ERP systems.
- Monitored machine data flow and identified issues when data was not received correctly.
- Analyzed and resolved communication errors to ensure stable and continuous data transfer.
- Implemented M-Code based logging (start log and end log) in CNC programs to capture part count and cycle time.
- Modified machine programs by inserting logging codes at the start and end of execution.
- Handled machine compatibility issues where certain CNC machines did not support specific M-Codes by providing alternative solutions.

- Resolved machine controller errors during program execution through debugging and iterative testing.
- Performed multiple rounds of testing and bug fixing to ensure accurate data capture in the ERP database.
- Ensured reliable data transfer through validation, error handling, and optimized communication setup.
- Collaborated with cross-functional teams to deploy and test the solution in a real-time manufacturing environment.

2.CNC Program Upload and Download System ,Veltech Auto Part India Private Limited

- Developed a web-based application using C# to upload and download CNC/Fanuc machine programs directly from a PC, eliminating manual pen drive transfers.
- Enabled seamless program transfer to both Ethernet-enabled and RS232-enabled machines.
- Implemented IP-based communication for Ethernet-enabled machines without requiring any machine-side modifications.
- Designed RS232 communication workflow for legacy machines, including necessary machine-side configuration setup.
- Built backend logic to handle reliable file transfer between PC and machines.
- Enhanced operational efficiency by reducing manual intervention and saving time in program deployment.
- Centralized machine program management through a user-friendly web interface.
- Faced and resolved challenges in RS232 communication where machines did not always accept the complete program during transfer.
- Identified issues such as partial data transfer and communication interruptions, and implemented fixes to ensure complete and accurate program upload/download.
- Analyzed root causes of transmission errors and optimized data transfer logic for better reliability.
- Handled differences between Ethernet and RS232 machines by applying appropriate communication strategies for each type.
- Ensured stable communication through proper error handling, validation, and repeated testing.

3.Zentroo ERP Website Development ,Freedom software solution Private Limited

Live Website: Zentroo.com

- Designed and developed a modern, responsive ERP product website for Zentroo to showcase manufacturing ERP solutions and business offerings.
- Built and maintained multiple pages including Home, Features, Benefits, Industries, Pricing, Blog, About, Support, and Contact pages.
- Implemented responsive layouts optimized for mobile, tablet, and desktop devices to ensure a seamless user experience across all screen sizes.
- Enhanced website UI/UX with modern design principles, interactive components, animations, and smooth navigation.
- Developed industry-specific solution pages to improve product visibility and customer engagement.
- Optimized website performance, page structure, and SEO readiness for improved search engine visibility.
- Integrated lead generation and contact inquiry forms to support customer acquisition and business growth.
- Collaborated with stakeholders to continuously improve website design, functionality, and user experience based on business requirements.

EDUCATION

B.Tech in Artificial Intelligence and Data Science ,Anna University, Tamil Nadu 2025

CERTIFICATION

Full Stack Web Development – Udemy